

Regression:

Example:

A company located in a certain region makes deliveries. The CEO of the company wants to study the relationship between the distance a shipment must travel and the length of time, in days, it takes the shipment to arrive at its destination. To investigate, a random sample of 10 shipments made last month is selected. The results are as follows:

Shipment	Distance (miles)	Shipping time (days)
1	656	5
2	853	14
3	646	6
4	783	11
5	610	8
6	841	10
7	785	9
8	639	9
9	762	10
10	762	9

a- Develop an estimated Regression equation to Predict shipping time based on Distance.

b- Obtain unexplained & total variation. Also

c- Find coefficient of determination
Find correlation also

Sol:

Let 'x' be the independent variable (distance).

Let 'y' be the dependent variable (Shipping time)

a- estimated regression equation is

$$\hat{y} = a + bX$$

X	Y	X ²	Y ²
3280	25	430336	25
11942	196	727609	196
3876	36	417316	36
8613	121	613089	121
4880	64	372100	64
8410	100	707281	100
7065	81	616225	81
5751	81	408321	81
7620	100	580644	100
6858	81	580644	81

$$\sum XY = 68295 \quad \sum X^2 = 5453565 \quad \sum Y^2 = 885$$

$$\Rightarrow b = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2}$$

$$= \frac{10(68295) - (7337)(91)}{10(5453565) - (7337)^2}$$

$$= \frac{15283}{704081} = 0.0217$$

$$a = \bar{Y} - b\bar{X}$$

$$\bar{Y} = \sum Y / n = 91/10 = 9.1$$

$$\bar{X} = \sum X / n = \frac{7337}{10} = 733.7$$

$$a = 9.1 - (0.0217)(733.7)$$

$$= 9.1 - 15.9213$$

$$a = -6.82$$

$$\hat{y} = -6.82 + 0.0217X$$

b- unexplained variation: $\sum$ total variation:

$$\begin{aligned}\text{unexplained variation} &= \sum (Y - \hat{Y})^2 \\ &= \sum Y^2 - a \sum Y - b \sum XY \\ &= 885 - (-6.82)(91) - \\ &\quad 0.0217(68295) \\ &= 1505.62 - 1482.0015 \\ &= 23.6185\end{aligned}$$

$$\begin{aligned}\text{Total variation: } \sum (Y - \bar{Y})^2 &= \sum Y^2 - \frac{(\sum Y)^2}{n} \\ &= 885 - \frac{(91)^2}{10} \\ &= 56.9\end{aligned}$$

Coefficient of determination:

$$R^2 = 1 - \frac{\text{unexplained variation}}{\text{Total variation.}}$$

$$\begin{aligned}&= 1 - \frac{\sum (Y - \hat{Y})^2}{\sum (Y - \bar{Y})^2} = 1 - \frac{23.6185}{56.9} \\ &= 0.58 = 58\% \quad \left[\begin{aligned} R &= \sqrt{0.58} = 0.7616 \\ R &= \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}} \end{aligned} \right]\end{aligned}$$

58% of variability in dependent variable (time) is explained due to its relationship with independent variable (distance).

Multiple Regression

$$Y = a + b_1 X_1 + b_2 X_2 + \dots + b_k X_k.$$

$b_1, b_2 \dots b_k$ (Regression coefficients)

$X_1, X_2 \dots X_k$ (Independent variables)

Y (dependent variables).

Prediction of CO₂ emission based on
based on engine size & number
of cylinders in a car.

It means guessing how much
CO₂ (carbon dioxide) a car will
Produce by looking at two things
(size of its engine & no of
Cylinders).

How much pollution a car
might make based on type
of engine. It has & how
big it is.